

```
karin@Karin's_Laptop CLANG64 /c/Users/karin/Music/CSE-3150/lab11
$ ./a.exe
Match value: 0
Mismatch: 200 100
Mismatch: 200 100
```

code:

```
#ifndef EC_TRADING_NEWS_H
#define EC_TRADING_NEWS_H

#include <vector>
#include <algorithm>

class ECTrendingNews;

// Subscriber interface. DON'T CHANGE THIS
class ECTSubscriber
{
public:
    virtual ~ECTSubscriber() {}
    virtual void Update() = 0;
    virtual int ContractCost() const = 0;
};

// pay-per click subscriber
class ECPayPerClickSubscriber : public ECTSubscriber
{
public:
    ECPayPerClickSubscriber(ECTrendingNews &news);
    virtual void Update() override {}
    virtual int ContractCost() const override { return 0; }
};

// contract subscriber
class ECContractSubscriber : public ECTSubscriber
{
public:
    ECContractSubscriber(ECTrendingNews &news);
    virtual void Update() override {}
    virtual int ContractCost() const override { return 100; }
};

class ECTrendingNews
{
public:
    ECTrendingNews();
    ~ECTrendingNews();

    void Subscribe(ECTSubscriber *pSub);
    void UnSubscribe(ECTSubscriber *pSub);

    void Notify();
    void ReceivePayment(int amount);
    int GetTotRevenue() const;

private:
    std::vector<ECTSubscriber*> subs;
    int revenue;
};

#endif
```

```

#include "ECTrendingNews.h"

ECPayPerClickSubscriber::ECPayPerClickSubscriber(ECTrendingNews &news)
{
    news.Subscribe(this);
}

ECContractSubscriber::ECContractSubscriber(ECTrendingNews &news)
{
    news.Subscribe(this);
}

ECTrendingNews::ECTrendingNews() : revenue(0) {}

ECTrendingNews::~ECTrendingNews()
{
    for (auto *p : subs) delete p;
}

void ECTrendingNews::Subscribe(ECSubscriber *pSub)
{
    subs.push_back(pSub);

    int cost = pSub->ContractCost();
    if (cost > 0)
        ReceivePayment(cost);
}

void ECTrendingNews::UnSubscribe(ECSubscriber *pSub)
{
    subs.erase(std::remove(subs.begin(), subs.end(), pSub), subs.end());
}

void ECTrendingNews::Notify()
{
    for (auto *p : subs)
    {
        p->Update();
        if (p->ContractCost() == 0)
            ReceivePayment(1);
    }
}

void ECTrendingNews::ReceivePayment(int amount)
{
    revenue += amount;
}

int ECTrendingNews::GetTotRevenue() const
{
    return revenue;
}

```